

The Role of Attention and Cognitive Workload in Measuring Levels of Task Complexity within Virtual Environments

Yobbahim J. Vite* Yaoping Hu*

Dept. of Electrical and Software Engineering
Schulich School of Engineering
University of Calgary, CANADA

ABSTRACT

This paper studied the role of attention and cognitive workload (CWL) in measuring levels of task complexity. Within a virtual environment (VE), participants undertook a task with a baseline and 3 levels of the complexity. The baseline and levels were defined as those in an existing work. A well-known ratio of attention, AR, was computed from recorded brainwaves of the participants during the task. Their CWL was derived from a de-facto questionnaire of the NASA task load index. The study's outcomes indicated that, while the role of the AR in the measurement remained unclear, the CWL could serve the measurement.

Keywords: VR, attention ratio, CWL, task complexity, EEG.

Index Terms: H.1.2 [User/Machine Systems]: Human information processing; H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities; H.5.2 [User Interfaces]: Haptic I/O

1 INTRODUCTION

In the daily life, a human individual can direct his/her attention to execute a task. The complexity of the task (task complexity) might be dependent of a situation, in which the task is executed. Most research efforts attempt to elucidate relationships between task performance of an individual and his/her cognitive workload (CWL) [1]. However, the quantification of the task complexity remains largely underexplored. A recent work has described a parametric definition of the task complexity and explored the relationship between the complexity and the individual's engagement in a task within a computer-generated virtual environment (VE) [2]. Based on brainwaves captured by an electroencephalogram (EEG), the engagement has been evaluated by using a well-known engagement ratio (ER) [3]. This existing work has mapped the ER with the levels of the task complexity, but considered the attention (a component of the engagement [4]) and the CWL.

There are many studies on how sustaining attention effects task performance. For example, empirical observations have shown that maintaining attention in a repetitive and monotonous task is much harder than in a mentally demanding and interesting one [5]. The observations have also indicated that vigilant attention – a process of maintaining consciousness on the task – lasts about 10.00 s. This process is related to stimuli discrimination for cognitive and motor responses (e.g., haptic feedback) by excluding high-level cognition (like spatial orienting and switching between tasks). However, the observations reveal no effect of the CWL.

Fortunately, a recent study has established a correlation between the CWL and the familiarity of information presentation for

completing tasks [6]. The study has found that the CWL is relatively constant for individuals familiarized with the mode of the presentation. In contrast, their CWL spikes drastically when such familiarization is absent – implying an increase in task complexity for them. That is, the CWL could indicate the task complexity.

One common way of measuring the attention to a task utilizes a ratio derived from the EEG brainwaves of an individual [3]. This ratio, called as AR, is the power spectral density (PSD) of the α band (9.00 ~ 12.00 Hz) over that of the β band (13.00 ~ 30.00 Hz) of the brainwaves as [3]:

$$AR = PSD_{\alpha}/PSD_{\beta} \quad (1)$$

Regarding the CWL, its de-facto measurement is the questionnaire of the NASA task load index (NASA-TLX). The CWL is the score summation of 6 weighted factors including mental demand, physical demand, temporal demand, performance, effort, and frustration. To the best of our knowledge, most studies have focused on investigating how individuals use the attention for their tasks. However, there is a gap of considering the attention and CWL for relating to a level of the task complexity. To fill this gap, we undertook a study by replicating the task and the parametric definition of the task complexity reported in the existing work [2].

2 METHODS

VE setup: A head-mounted device (Pimax; Pimax Technology, China) was utilized to display a visual scene, which consisted of a pipe on a contrasted background. The pipe was replicated from its description in the existing work [2]. The task of a human participant was to push an invisible ball through the pipe by using the stylus tip of a haptic device (TOUCH; 3D Systems, USA). The levels of the task complexity were 4 folds as: baseline (B = 0.00), level 1 (L1 = 10.64), level 2 (L2 = 22.97), and level 3 (L3 = 35.59). All the levels were derived from Eq. (2) of the existing work [2].

As depicted in Fig. 1, each participant wore the cap of an EEG (Enobio; Neuroelectrics, Spain) with a 10-20 placement of its 20 electrodes. Two EEG electrodes, FP1 and FP2, were removed to reduce pressure from the Pimax to the forehead of the participant for his/her comfort. The other 18 electrodes were placed on his/her scalp to record EEG signals at 500.00 Hz.

The participant started the task by moving the stylus tip of the TOUCH device to click a cube at the left end of the pipe, and then articulated the tip to push the ball from the right end to the left end of the pipe. Force feedback at 0.8 N was delivered by the stylus to his/her right hand, when the ball collided with the internal wall of the pipe. The feedback was larger than 0.45 N required for human perception of a force [7].

Participants: With an informed consent, all 10 right-handed participants (aged at 26.7 ± 1.7 years) underwent a pre-test to confirm their normal color vision and at least 40' stereo acuity. They were unaware of the purpose of the study. The number of participants was computed by applying the Lehr's formula [8]. An ethics clearance was granted by our institute for the study.

* email: {yobbahimjavierisreal, huy}@ucalgary.ca



Figure 1: The VE setup of the study.

Procedure: The participant took a seat in front of a monitor while holding the stylus of the TOUCH device. After confirming his/her comfort, the participant underwent the recording of EEG signals in a resting state with the eyes closed and then open. A baseline session corresponding to the level B of the task complexity lasted 3.0 minutes, followed by the administration of the NASA-TLX questionnaire and a learning session of 5.0 minutes. The learning ensured that the participant could perform the task stably with the number of ball-wall collisions below 40. After then, a test session was carried out. The participant executed 10 trials per level of the task complexity for a total of 30 trials. The order of the levels was always from L1, L2, to L3. The participant spent 30.00 ~ 45.00 minutes in the test session. His/her EEG signals and performance (e.g., the number of collisions and completed trials, and the temporal length of a trial) were collected. The NASA-TLX questionnaire was administered after completing each level.

The EEG signals of the recorded electrodes were filtered to remove artifacts at 60Hz and from muscular impulses. The Welch method was then applied to compute PSD of a frequency band over a 2.0-second window. All resulted PSD were used to calculate the AR and ER accordingly. The data collected via the NASA-TLX questionnaires were to yield the CWL of each level. An ANOVA analysis was carried out for each data, after its distribution passed a normality test using a Kolmogorov-Smirnov approach. Significant differences derived from ANOVA analyses were further evaluated through Bonferroni post-hoc tests.

3 RESULTS AND DISCUSSION

On average, each participant carried out 35 trials for all levels including B. As depicted in Fig. 2, the AR trended decreasingly from B to L3 in contrast to the increase of the task complexity. This AR trend corresponded inversely to the trend of the ER. After confirming that the AR of each level passed a normality test, an ANOVA analysis of the AR indicated a significant difference among all levels [$F(2,27) = 89.11; p < 0.001$]. Post-hoc analyses of the AR showed that all pairs are significant except the L1-L2 pair, as illustrated in Fig. 2. Again, this exception might reflect the peculiarity of L2 as reported in [2].

The AR trend in opposition of the task complexity could denote a relation with the average time of trials at each level. In general, the more difficult the task was, the shorter time the participants spent on trials. This counter-intuitive observation might be a consequence of learning, because each participant performed his/her task following the fixed order of the levels from the easiest B to the most difficult L3. Interestingly, the average time of trials was about 10.00 s long, at the similar range that vigilant attention could be triggered [5]. Showing an overall increase in performance from B to L3, all participants made fewer contacts and completed more trials from the easiest to most difficult levels. The performance corresponded in a similar way to that observed in [2].

The CWL derived from the NASA-TLX data increased from B to L3 except L2, as shown in Fig. 3. This observation reflected the participants' perceived task difficulty at each level. After confirming the normality distribution of the CWL, an ANOVA analysis revealed a significance difference among all levels [$F(3,36) = 5.56; p < 0.005$]. Post-hoc analyses indicated a significant

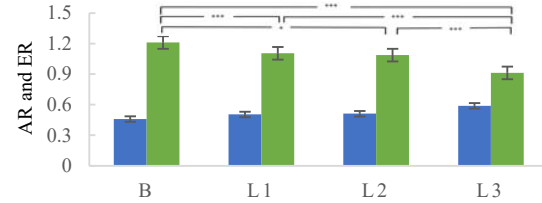


Figure 2: Comparison of averaged AR (green) and ER (blue) at each level of the task complexity. [Error bars are standard errors. Symbols ***, **, and * denote Bonferroni differences with $p < 0.001$, $p < 0.01$, and $p < 0.05$.]

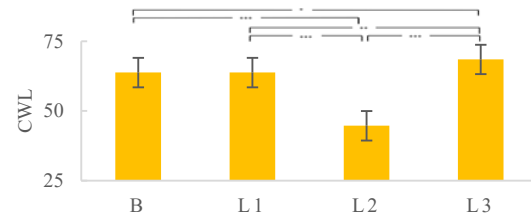


Figure 3: Average CWL at each level of the task complexity. [Error bars are standard errors. The symbols of ***, **, and * denote Bonferroni differences with $p < 0.001$, $p < 0.01$, and $p < 0.05$.]

difference between each level pair except the B-L1 pair, as depicted in Fig. 3. Moreover, the CWL of L2 was much smaller than that of other levels. This peculiarity of L2 might mirror its failed normality test of engagement ration (ER) reported in [2]. This mirroring could imply that L2 was unsuitable for the participants to distinguish from other levels and, thus, needs further investigations.

4 CONCLUSION

Our observations imply that the AR might be unsuitable to describe the task complexity, but could be a secondary measure for monitoring the readiness of executing a task. However, the CWL could measure the task complexity. Future work needs to examine the effect of a trial length beyond 10.0 s related to the attention.

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